EEE3096S Practical 3 — Voltage Mirror

Student report

Gil Sharon (SHRGIL004); Tri Tac Le (LXXTRI004)
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1. Measurements and predictions

Measured analogue reference: $V_R = 3.292$ V.
DMM range/resolution: not recorded. Voltages below are reported to 0.001 V; this does not establish the instrument's resolution or accuracy.
Nominal ADC/DAC interval: $V_R/4096 = 0.8037109375$ mV.
Predicted coarse output step: $V_R/64 = 51.4375$ mV.

The nominal DAC prediction is $V_{\mathrm{PA4,pred}} = DV_R/4096$. The measured PA5 reading for each input setting is used for both modes.

Target (V)	Mode	Measured PA5 (V)	C^*	D^*	Predicted PA4 (V)	Measured PA4 (V)
0.800	FULL12	0.806	994	994	0.798889	0.802
0.800	COARSE6	0.806	993	960	0.771563	0.775
1.650	FULL12	1.654	2058	2058	1.654037	1.656
1.650	COARSE6	1.654	2060	2048	1.646000	1.645
2.500	FULL12	2.508	3118	3118	2.505971	2.511
2.500	COARSE6	2.508	3115	3072	2.469000	2.471

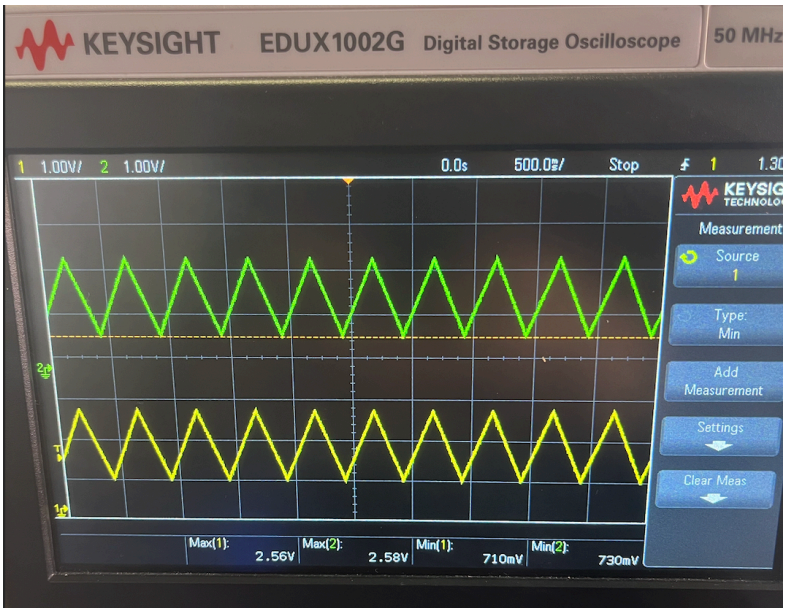
*Representative codes selected from the recorded ranges, rather than simultaneously recorded LCD pairs. FULL12 uses the midpoint of the overlap of the recorded C and D ranges, with $D = C$. COARSE6 uses a midpoint C (half-integers rounded down) and the observed fixed D. No selection was made to minimise voltage error.

Original ranges, in ascending target order: FULL12 C: 990–998, 2052–2064, 3110–3121; D: 990–999, 2049–2064, 3115–3125. COARSE6 C: 990–996, 2055–2066, 3111–3120; D: 960, 2048, 3072.

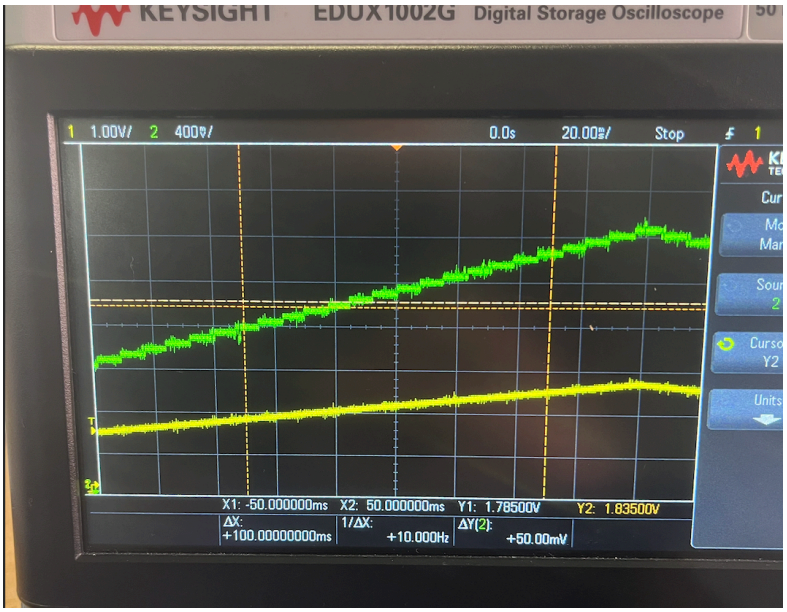
Generator settings: triangle (ramp, 50% symmetry), 2 Hz; 1.7 V_{pp}; 1.65 V offset.
PA5 minimum/maximum: approximately 0.710/2.56 V
(Capture 1, CH1 automatic measurements).
Measured adjacent coarse step: 50.00 mV.
Measured minus predicted: $50.00 - 51.4375 = -1.4375$ mV.
Percentage difference: $100(50.00 - 51.4375)/51.4375 = -2.795\%$.

2. Two real scope captures

In both captures, CH1 (yellow) is PA5 input and CH2 (green) is PA4 output. The traces are vertically displaced for visibility.



Capture 1 — FULL12. Both channels: 1.00 V/div; time scale: 500 ms/div. The supplied capture shows approximately ten cycles, rather than the requested single cycle. Both traces appear triangular; individual 12-bit steps are unresolved at this scale.



Capture 2 — COARSE6. CH1: 1.00 V/div; CH2: 400 mV/div; time scale: 20.00 ms/div. Adjacent output plateaus are visible. CH2 voltage cursors read $Y_1 = 1.78500$ V and $Y_2 = 1.83500$ V, giving $\Delta Y = 50.00$ mV. The horizontal cursor separation is not used in the step calculation.

3. Explanation

Mapping. For an ADC code $C \in \{0, \ldots, 4095\}$, FULL12 uses $D = C$, whereas COARSE6 uses

$$D = 64 \left\lfloor \frac{C}{64} \right\rfloor .$$

The coarse commands are $0, 64, 128, \ldots, 4032$: 64 distinct commands spaced by 64 DAC counts. The largest is $64\lfloor 4095/64 \rfloor = 64(63) = 4032$; 4096 is outside the 12-bit code range. The nominal output spacing is $64V_R/4096 = V_R/64 = 51.4375$ mV. The recorded coarse command values agree with this mapping throughout their respective ADC ranges. The separate FULL12 ranges do not verify equality for an individual sample; the representative pairs in the table impose the intended relationship.

Measured steps. The cursors in Capture 2 give a coarse step of 50.00 mV, which is 1.4375 mV below the nominal prediction, a signed difference of -2.795%. At 400 mV/div this step occupies only 0.125 vertical divisions, so cursor placement and the visible trace noise limit the measurement. The displayed decimal places do not establish that level of accuracy. Scope vertical accuracy, reference-voltage uncertainty and DAC gain, offset or nonlinearity may also contribute. Without instrument specifications and repeated measurements, the difference cannot be attributed to one cause or declared statistically significant.

Resolution and accuracy. FULL12 retains all 12 ADC bits and commands the DAC in nominal 0.803711 mV increments. Its smooth-looking trace does not establish continuous voltage resolution: the discrete steps are too small to distinguish at the displayed scale. COARSE6 discards the lower six code bits, making the staircase visible, while both hardware converters remain configured for 12 bits. Resolution describes the available code spacing; accuracy describes closeness to the physical voltage. Quantisation, reference error, noise, converter offset and gain errors, nonlinearity and loading can affect accuracy even with 12-bit resolution.

One DC result. For the FULL12 input measured as 1.654 V, the representative pair $C = D = 2058$ gives

$$\frac{CV_R}{4096} = \frac{DV_R}{4096} = 1.654037109375 \text{ V}.$$

This ADC estimate exceeds the measured input by about 0.037 mV, while the measured output of 1.656 V exceeds the nominal DAC prediction by about 1.963 mV. The actual measured output is 2 mV above the input. The very small input difference is below the reporting increment and should not be interpreted as demonstrated accuracy. These calculations use a representative code, not a simultaneous measurement. Copying the code preserves the digital value but does not cancel the separate ADC and DAC errors, noise, quantisation or measurement uncertainty; it therefore does not guarantee identical physical input and output voltages.